ABHIRAM KALATHUR

Wichita, KS | +1 (770) 570-0483 | kalathurabhiram@outlook.com | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Manufacturing and Industrial Engineering professional with 2+ years of experience across aerospace manufacturing and high-volume operations, supporting production readiness, process improvement, design review, and shop-floor execution. Experienced in working with Manufacturing, Design, and Quality teams to resolve production issues, evaluate manufacturability, and strengthen process consistency from engineering release through production. Brings a practical background in Lean manufacturing, production analysis, mechanical design support, and continuous improvement, with a strong focus on efficient, reliable, and manufacturable outcomes.

TECHNICAL SKILLS

Manufacturing & Process Engineering: Manufacturing Process Development, Production Planning, Production Readiness, Process Mapping, Process Flow Analysis, Time Studies, Line Balancing, Cycle Time Analysis, Capacity Analysis, Bottleneck Analysis, Process Optimization, Standardized Work

Mechanical Design & CAD: CATIA V5/V6, SolidWorks, Creo (Exposure), AutoCAD (Exposure), 3DEXPERIENCE, GD&T, Engineering Drawings, Tolerance Analysis, DFM, Producibility Analysis

Lean & Quality: Lean Manufacturing, Continuous Improvement, 6S, Root Cause Analysis, 5 Whys, FMEA, Corrective Actions, Non-Conformance Investigation, Process Validation, First-Pass Quality

Production Analytics: Advanced Excel, Production KPI Analysis, Cycle-Time Tracking, Build Status Tracking, Trend Analysis, Process Performance Analysis

Engineering Systems: PLM, Engineering Change Review, Configuration Control, Work Instructions, Manufacturing Documentation

Manufacturing Simulation: RoboDK, ABB IRB 4400/60, Robotic Work-Cell Simulation, Material Handling Simulation, Motion-Path Optimization

AI-Assisted Engineering: Microsoft Copilot, AI-Assisted Technical Documentation, AI-Assisted Process Analysis

PROFESSIONAL EXPERIENCE

Manufacturing Engineer (Research Associate) **Dec 2024 - Present**

National Institute for Aviation Research (NIAR) | Wichita, KS

- Streamlined component flow through time studies, line balancing, cycle-time analysis, and standardized work, cutting processing time by 40% while creating a more balanced production sequence.
- Built Advanced Excel trackers for build status, assembly progress, cycle time, and production KPIs, giving engineering teams clearer visibility into bottlenecks and more reliable inputs for production planning.
- Examined dimensional and yield data from pilot runs of 20-50 units, combining tolerance analysis with process validation to trace defect sources and strengthen first-pass production quality.
- Eliminated recurring manufacturing rework through 5 Whys, Root Cause Analysis, and corrective-action reviews, bringing engineering rework down to 10-20% through more consistent validation practices.
- Prepared production readiness for 5-8 new part numbers per quarter by reviewing process flow, capacity needs, manufacturing requirements, and work instructions before shop-floor release.
- Verified CATIA V5/V6 and 3DEXPERIENCE models, GD&T, engineering drawings, and PLM revisions using DFM and producibility analysis, catching tolerance and manufacturing risks before fabrication or assembly.
- Partnered with Design and Quality Engineering to close non-conformances through configuration control, process validation, and engineering-change review, helping maintain product conformity and dependable production schedules.
- Applied Microsoft Copilot to technical documentation and process analysis, refining work instructions and engineering summaries so manufacturing findings were easier to review, communicate, and reuse across teams.

Operations Warehouse Associate | Lean & Process Improvement **Jul 2023 - Dec 2023**

Amazon | Newcastle upon Tyne, UK

- Inspected outbound packages against quality standards and work instructions, catching labeling and packaging errors before dispatch and helping maintain accurate order fulfillment.
- Coordinated material movement across picking, packing, and outbound handoffs, helping associates keep orders moving efficiently and limiting delays during high-volume shifts.
- Applied Lean Manufacturing and 6S practices to organize work areas and standardize material placement, decreasing unnecessary handling and making daily workflows easier to execute.
- Resolved recurring process deviations through structured quality checks and escalation procedures, limiting downstream errors and avoiding repeat handling of non-conforming orders.
- Lowered rework by 15% through clearer task sequencing and consistent process adherence, helping associates complete fulfillment activities with fewer corrective steps.
- Monitored order flow and operational handoffs during daily fulfillment activities, surfacing recurring delays to operations leads and supporting steadier outbound execution.

- Reinforced safety, quality, and standard operating procedures with team members, strengthening process discipline and consistency across routine warehouse activities.
- Partnered with warehouse associates and operations leads to address quality exceptions and flow disruptions, restoring smoother order movement and supporting dependable outbound performance.

PROJECTS

Robot Programming & Material Handling Automation

- Configured a RoboDK work cell with an ABB IRB 4400/60 robot for a 4 × 4 pallet layout, programming pick-and-place sequences to establish repeatable material handling.
- Simulated robot travel, palletizing logic, and cycle-time behavior in RoboDK, tracing excess motion to repeated home-position transitions that slowed the handling sequence.
- Refined motion paths and transition logic to shorten non-value-added travel, demonstrating a more efficient robotic workflow before physical implementation.

Global Grab Productions - Manufacturing Business Simulation

- Modeled an assembly system with Lean Manufacturing, SIPOC, process-flow analysis, and capacity analysis to expose sequencing constraints affecting throughput and delivery performance.
- Prioritized high-risk failure modes through FMEA and corrective-action analysis, targeting defect-driven rework that disrupted assembly flow and process consistency.
- Raised simulated on-time delivery from 85% to 95% by addressing critical failure modes and aligning corrective actions with production flow and capacity requirements.

Mechanical Component DFM & Tolerance Design

- Modeled a mechanical component and assembly in SolidWorks using GD&T and tolerance analysis, defining manufacturable geometry and drawing requirements for repeatable production.
- Evaluated design features through DFM and producibility analysis, revising tolerances and component interfaces to simplify fabrication while preserving functional requirements.
- Documented the finalized design with engineering drawings and CAD revisions, creating a clearer manufacturing package for design review and downstream production use.

EDUCATION

Master of Science in Engineering Management (Industrial & Manufacturing) **Jan 2024 - Dec 2025**
Wichita State University | Wichita, KS

Master of Science in Mechanical Engineering with Advanced Practice **Jan 2020 - Dec 2022**
Northumbria University | Newcastle upon Tyne, UK

Bachelor of Technology in Mechanical Engineering **Jun 2014 - May 2018**
Raghu Institute of Technology | Andhra Pradesh, India

CERTIFICATIONS

- SolidWorks xDesign Associate Certification
- SME Lean Bronze Knowledge Certification
- Six Sigma Principles - **Coursera**
- Introduction to Operations Management - **Coursera**
- Digital Manufacturing & Design Technology Specialization - **Coursera**
- Lean Six Sigma Foundations - **LinkedIn Learning**